

Fibrewise Characterization of Resolution Geometry

Fixed-Protocol Representation, Maximal Admissibility, and a Finite Uniqueness Attack

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Abstract

Resolution Geometry (RG) is formulated as a modular geometry of distinctions supported by a declared observation system. Its primitive input is a complete statistical experiment $\mathcal{E}$ together with an explicit admissibility protocol Γ . Equality of complete observation laws induces the observational quotient $Q_{\mathcal{E}}$, and observational extensionality gives its universal factorization property. In the regular finite classical module, sufficient-Markov invariance characterizes Fisher–Rao distinguishability up to declared normalization. A comparison metric G and threshold τ then determine the unique hard resolved projector $P_{\tau} = \mathbf{1}_{(\tau, \infty)}(G^{-1}F)$. The final sector is fixed by the missing non-arbitrariness principle: it is the greatest resolved subobject satisfying every declared nuisance, stability, conditioning, coarse-graining, and transport gate. Naturality and cocycle laws transport and glue these local objects.

The resulting representation theorem is fibrewise: for fixed $(\mathcal{E}, \Gamma)$, the object

$$\mathrm{RG}(\mathcal{E}, \Gamma) = (Q_{\mathcal{E}}, g_F, P_{\tau}, W_{\mathrm{obs|adm}}, \{T_{ij}\}, \Gamma)$$

is unique up to protocol-preserving natural isomorphism and the declared Fisher normalization. The proof is an assembly of established mathematics: quotient universal properties, statistical sufficiency and experiment comparison, a Čencov-type metric characterization, spectral functional calculus, greatest subobjects in finite subspace lattices, and standard gluing. A historical theorem chart makes this dependency explicit. Importantly, admissibility of a decision rule does not by itself create or uniquely select a sufficient reduction; minimal sufficiency belongs to the experiment, while decision-theoretic admissibility constrains rules relative to a loss. RG’s uniqueness therefore comes from the complete bundled protocol, not from admissibility alone. Entropy decomposes downstream along quotient fibres by Shannon’s chain rule. Finite exhaustive attacks isolate admissibility completeness as necessary and leave one survivor in the stated hard-spectral class. No protocol-free uniqueness, latent ontology, unrestricted infinite-dimensional extension, or reduction of physics to information is claimed.

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1 Introduction

The foundational question is not whether every conceivable observation theory must equal Resolution Geometry. In an unrestricted class, no such uniqueness is plausible: an experiment does not determine a comparison scale, decision threshold, nuisance convention, stability class, or connection. The correct question is conditional:

$$(\mathcal{E}, \Gamma) \longmapsto \text{RG}(\mathcal{E}, \Gamma). \quad (1)$$

Here $\mathcal{E}$ declares the complete observation law and Γ declares the operational contract under which a direction counts as admissibly observable.

This shift is analogous to familiar geometric constructions. A manifold does not select a Riemannian metric, but a Riemannian manifold (M, g) determines its Levi–Civita connection from metric compatibility and vanishing torsion. Likewise, RG does not claim that $\mathcal{E}$ selects Γ . It claims that the bundled input $(\mathcal{E}, \Gamma)$ determines the observable quotient, distinguishability geometry, hard resolution, maximal admissible sector, and compatible transport under explicit hypotheses.

The earlier modular axioms already characterized the quotient, the regular classical metric module, the hard resolved projector, and natural gluing. The remaining finite-dimensional ambiguity was that admissibility had been stated only as containment,

$$W_{\text{obs|adm}} \subseteq \text{im } P_\tau \cap N^\perp \cap S \cap C \cap T, \quad (2)$$

which permits the zero sector and arbitrary strict subspaces. The finite attack reported here isolates the missing condition: admissibility must be complete as well as sound. Thus $W_{\text{obs|adm}}$ is the maximal resolved subobject satisfying all declared gates.

Contribution. The paper provides:

1. an exact definition of the fixed protocol fibre;
2. an admissibility soundness/completeness pair;
3. a proof of maximal-sector uniqueness in finite-dimensional linear fibres;
4. a modular fixed-protocol representation theorem;
5. a natural-isomorphism corollary across transported presentations;
6. an exhaustive finite counterexample search and axiom-independence witness;
7. explicit non-claims separating proved finite results from open extensions.

2 Established mathematical grounding

The result is an assembly theorem. Its load-bearing modules are established constructions used within explicitly stated domains.

2.1 Quotients and universal properties

For a complete statistical experiment $\mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta})$, equality of laws defines

$$\theta \sim_{\mathcal{E}} \theta' \iff P_\theta = P_{\theta'}. \quad (3)$$

The quotient $Q_{\mathcal{E}} = \Theta / \sim_{\mathcal{E}}$ has the standard factorization property for maps constant on equivalence classes. This is an ordinary quotient universal property; universal constructions are unique up to unique compatible isomorphism. The claim made here is no stronger than that standard fact at the quotient layer.

2.2 Fisher–Rao characterization

In the regular finite classical setting, a Čencov-type theorem characterizes the Fisher information metric, up to a positive scalar normalization, among smooth metric families invariant under sufficient Markov embeddings. Modern extensions establish related formulations for exponential families under explicit invariance hypotheses. RG imports this as a metric module; observational quotient logic alone does not select Fisher geometry.

2.3 Spectral functional calculus

For a positive comparison metric G and positive-semidefinite Fisher tensor F , the information operator

$$L = G^{-1}F \quad (4)$$

is self-adjoint in the G inner product. The finite-dimensional spectral theorem supplies a G -orthonormal eigenbasis, and Borel functional calculus supplies the canonical spectral projection

$$P_\tau = \mathbf{1}_{(\tau, \infty)}(L). \quad (5)$$

The projector is therefore not an extra coordinate after (F, G, τ) is fixed.

2.4 Maximal subobjects

The subspaces of a finite-dimensional vector space form a complete lattice: meets are intersections and joins are linear spans. Consequently, for a finite family of declared linear admissibility subspaces, their intersection is the unique greatest subspace contained in every member. This elementary lattice fact is the established grounding of admissibility completeness in the present theorem. In a more general category, the corresponding construction requires the relevant pullbacks or meets of subobjects; their existence is not assumed automatically outside the finite linear setting.

2.5 Blackwell order and information monotonicity

Blackwell comparison orders statistical experiments by informativeness: a less informative experiment is obtained by garbling a more informative one, equivalently under the classical decision-theoretic hypotheses it cannot improve every decision problem. RG uses only the compatible monotonicity requirement that parameter-independent processing cannot create distinguishability. Blackwell order does not choose G , τ , or the nuisance convention; it constrains admissible processing.

3 Established-theorem timeline and dependency ledger

The following chart is not a novelty claim. It records which established results carry each layer of the construction and prevents the assembled RG theorem from being mistaken for a new proof of its classical modules.

Table 1: Established results underlying the RG assembly.

Date	Result	Established force	Role in RG
1922	Fisher: sufficiency and statistical information	Introduces lossless statistical reduction relative to a model and the information carried by likelihood variation.	Historical basis for experiment-relative information; not itself a quotient-geometry theorem.
1945–1947	Rao–Blackwell theorem	Conditioning an estimator on a sufficient statistic cannot increase convex risk under the usual hypotheses.	Shows that sufficient reductions support essentially complete decision classes; it does not make admissibility select the statistic.
1948	Shannon chain rule	For a deterministic quotient $Q = \pi(X)$, $H(X) = H(Q) + H(X Q)$.	Exact observable/fibre entropy decomposition.
1950	Wald decision theory and complete-class results	Under explicit regularity, admissible procedures are represented within Bayes or generalized-Bayes complete classes.	Decision layer and risk geometry; no protocol-free observational quotient follows.

Date	Result	Established force	Role in RG
1951–1953	Blackwell comparison of experiments	Garbling orders experiments by decision informativeness; statistical equivalence preserves decision problems.	Information monotonicity and comparison of observable experiments.
1950s	Minimal sufficiency and Lehmann–Scheffé theory	Minimal sufficient statistics, when they exist in the stated category, are unique up to measurable equivalence; completeness yields uniqueness of unbiased estimators through the statistic.	Canonical statistical reduction under its own assumptions, distinct from rule admissibility.
1955	Basu theorem	A boundedly complete sufficient statistic is independent of every ancillary statistic under the theorem’s hypotheses.	Clarifies informative versus ancillary structure; does not erase latent variables.
1960s	Le Cam comparison and deficiency	Quantifies approximate equivalence and loss between experiments.	Natural route to approximate RG equivalence beyond exact isomorphism; not proved here.
1970s–1982	Čencov characterization	In the regular finite classical category, sufficient–Markov invariance characterizes Fisher–Rao geometry up to scale.	Metric module only; quotient logic alone cannot select Fisher.
Classical	Spectral theorem and functional calculus	A self-adjoint information operator has canonical spectral projectors.	Uniqueness of P_τ once (F, G, τ) is fixed.
Classical	Subspace lattices and categorical uniqueness	Intersections are greatest common subspaces; universal/greatest objects are unique by structure-preserving isomorphism.	Maximal admissible sector and fibrewise canonicity.

3.1 The essential correction to the decision-theory reading

The strongest defensible statement is not “admissibility forces a unique sufficient statistic.” Admissibility is defined only after an action space and loss are fixed, whereas sufficiency and minimal sufficiency are properties of the statistical experiment. A sufficient statistic permits Rao–Blackwell improvement for convex losses, and complete-class theorems relate admissible rules to Bayes procedures under regularity. Neither implication reverses into protocol-free selection of a unique observation geometry. RG closes this logical gap by declaring Γ and then imposing both admissibility soundness and admissibility completeness. The quotient is universal; the final sector is greatest only inside its fixed protocol fibre.

Proposition 3.1 (Separation of statistical and protocol uniqueness). *Let $\mathcal{E}$ be a statistical experiment and let Γ be a fixed admissibility protocol. Minimal sufficiency, when available, characterizes a reduction of $\mathcal{E}$ up to the relevant measurable equivalence. The RG sector*

$W_{\text{obs|adm}}$ is instead characterized as the greatest subobject satisfying the gates of Γ . Decision-rule admissibility alone implies neither construction.

Proof. Minimal sufficiency is ordered among sufficient statistics and contains no loss function or threshold. Decision admissibility is ordered by risk and therefore depends on the declared decision problem. The RG sector additionally depends on G , τ , nuisance, stability, conditioning, coarse-graining, and transport data. Since these inputs can vary while $\mathcal{E}$ remains fixed, no implication from decision admissibility alone can determine them. Once Γ is fixed, the greatest-subobject argument below gives uniqueness of $W_{\text{obs|adm}}$. $\square$

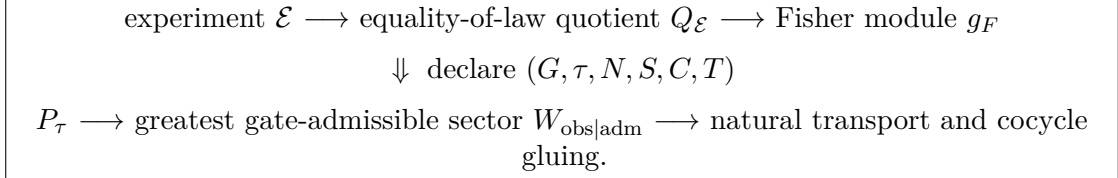


Figure 1: Logical dependency of the final RG object. Vertical declaration marks the point at which protocol-relative structure enters.

4 Declared admissible experiments

Definition 4.1 (Complete statistical experiment). *A complete statistical experiment is*

$$\mathcal{E} = (\Theta, \mathcal{O}, \{P_{\theta}\}_{\theta \in \Theta}), \quad (6)$$

where Θ is a parameter or preparation domain, $\mathcal{O}$ is an outcome space, and P_{θ} is the complete outcome law at θ .

Definition 4.2 (Admissibility protocol). *An admissibility protocol is a tuple*

$$\Gamma = (G, \tau, N, S, C, T), \quad (7)$$

where G is a covariant positive comparison structure, τ is a resolution threshold or threshold family, N is a nuisance/gauge convention, S is a stability protocol, C is a conditioning and coarse-graining protocol, and T is transport/gluing data.

Definition 4.3 (Declared admissible experiment). *The primitive RG object is the bundled pair*

$$\mathbb{E} = (\mathcal{E}, \Gamma). \quad (8)$$

No component of Γ is inferred silently from inaccessible latent labels, undeclared priors, or an external physical interpretation.

Definition 4.4 (Protocol fibre). *Fix a protocol Γ . The protocol fibre Exp_{Γ} is the category whose objects are experiments carrying that protocol type and whose morphisms are experiment morphisms together with the prescribed covariant transport of all protocol data. The target fibre ObsGeom_{Γ} consists of quotient geometries, metric modules, resolved projectors, admissible sectors, and transition maps preserving the fixed contract.*

Remark 4.5. “Fixed protocol” means fixed as structured data, not fixed coordinate arrays. Under a change of representation, G , nuisance data, gates, and transports move covariantly. Fibrewise uniqueness is therefore uniqueness up to protocol-preserving natural isomorphism, not literal equality of matrices.

5 Axioms for fixed-protocol observable geometry

Axiom 5.1 (Observational extensionality). *Every experiment-internal construction Φ satisfies*

$$P_\theta = P_{\theta'} \implies \Phi(\theta) = \Phi(\theta'). \quad (9)$$

Axiom 5.2 (Naturality). *Observable constructions commute with experiment isomorphisms and with covariant transport of the complete protocol.*

Axiom 5.3 (Information order). *A parameter-independent garbling cannot increase the local distinguishability tensor.*

Axiom 5.4 (Off-discriminant regularity). *Continuous structures vary continuously and discrete invariants remain locally constant away from the declared discriminant, including threshold crossings, rank changes, nuisance-rank changes, conditioning boundaries, stability failures, and transport singularities.*

Axiom 5.5 (Composition and gluing). *Local comparison maps preserve identities and composition. On overlaps,*

$$T_{ii} = \text{id}, \quad T_{jk}T_{ij} = T_{ik}. \quad (10)$$

Assumption 5.6 (Regular classical metric module). *On regular finite classical statistical models, local distinguishability is represented by a continuous smooth tensor family invariant under sufficient Markov embeddings, with fixed positive normalization.*

Axiom 5.7 (Hard spectral resolution). *The resolved selector P is G -self-adjoint, idempotent, commutes with $L = G^{-1}F$, agrees with the threshold mask on every generalized eigenspace, and composes on independent direct sums.*

Definition 5.8 (Gate-admissible resolved subobjects). *Let $R_\tau = \text{im } P_\tau$. For fixed Γ , let $\text{Adm}_\Gamma(R_\tau)$ be the set of resolved subobjects satisfying all declared nuisance, stability, conditioning, coarse-graining, and transport gates.*

Axiom 5.9 (Admissibility soundness). *The reported sector belongs to the admissible family:*

$$W_{\text{obs|adm}} \in \text{Adm}_\Gamma(R_\tau). \quad (11)$$

Equivalently, it contains no direction rejected by any declared gate.

Axiom 5.10 (Admissibility completeness). *The reported sector is greatest among gate-admissible resolved subobjects:*

$$U \in \text{Adm}_\Gamma(R_\tau) \implies U \subseteq W_{\text{obs|adm}}. \quad (12)$$

Thus $W_{\text{obs|adm}}$ retains every direction accepted by the complete declared protocol.

Remark 5.11. *Soundness is a safety condition; completeness is a non-arbitrariness condition. Soundness alone permits arbitrary strict reductions. Completeness forbids discarding an accepted direction without adding a new declared gate.*

6 Characterization ladder

Theorem 6.1 (Quotient representation). *Under observational extensionality, every experiment-internal map $\Phi : \Theta \rightarrow Z$ factors uniquely through $Q_\mathcal{E}$. There is a unique $\tilde{\Phi} : Q_\mathcal{E} \rightarrow Z$ such that*

$$\Phi = \tilde{\Phi} \circ \pi_\mathcal{E}. \quad (13)$$

Proof. Define $\tilde{\Phi}([\theta]) = \Phi(\theta)$. Extensionality makes the definition representative-independent. Surjectivity of $\pi_{\mathcal{E}}$ gives uniqueness. $\square$

Corollary 6.2 (Canonical observable domain). *Any two realizations satisfying the quotient universal property are uniquely isomorphic by an isomorphism commuting with the quotient maps.*

Proof. This is the standard uniqueness argument for universal quotients: each candidate factors uniquely through the other, and uniqueness forces the composites to be identities. $\square$

Theorem 6.3 (Classical metric module). *Under the regular classical metric assumptions, the local distinguishability metric is Fisher–Rao up to the fixed normalization.*

Proof. This is the imported Čencov characterization module. The present paper does not reprove it and uses it only in its stated domain. $\square$

Theorem 6.4 (Unique hard resolved projector). *Let $F \succeq 0$, $G \succ 0$, and $\tau \notin \text{spec}(G^{-1}F)$. The unique operator satisfying hard spectral resolution is*

$$P_{\tau} = \mathbf{1}_{(\tau, \infty)}(G^{-1}F). \quad (14)$$

Proof. The operator $L = G^{-1}F$ is G -self-adjoint and therefore admits a G -orthonormal spectral decomposition. Commutation makes each eigenspace invariant. Idempotence and G -self-adjointness restrict the selector on each eigenspace to an orthogonal projection. Threshold consistency fixes that projection to identity above τ and zero at or below τ . Functional calculus gives the displayed projector and direct-sum composition. $\square$

Lemma 6.5 (Intersection realization of a finite linear protocol). *Suppose the fixed gates of Γ are represented by linear subspaces $A_1, \dots, A_k \subseteq R_{\tau}$. Then*

$$W_* = \bigcap_{a=1}^k A_a \quad (15)$$

is the unique greatest subspace satisfying every gate.

Proof. The intersection is a subspace contained in every A_a , hence it is sound. If U is contained in every A_a , then by the definition of intersection $U \subseteq W_*$. Hence W_* is greatest. If W' is another greatest element, then $W_* \subseteq W'$ and $W' \subseteq W_*$, so $W' = W_*$. $\square$

Theorem 6.6 (Maximal admissible-sector characterization). *In a finite-dimensional linear protocol fibre whose declared gates admit a greatest common resolved subobject, soundness and completeness characterize a unique final sector:*

$$W_{\text{obs|adm}} = \max \text{Adm}_{\Gamma}(R_{\tau}). \quad (16)$$

For linear subspace gates this becomes

$$W_{\text{obs|adm}} = R_{\tau} \cap A_N \cap A_S \cap A_C \cap A_T. \quad (17)$$

Proof. Existence follows from the assumed greatest admissible subobject, or from the intersection lemma in the linear case. If W and W' both satisfy soundness and completeness, completeness of W applied to W' gives $W' \subseteq W$, while completeness of W' applied to W gives $W \subseteq W'$. Thus $W = W'$. $\square$

Corollary 6.7 (No undeclared shrinkage). *Any strict subobject $U \subsetneq W_{\text{obs|adm}}$ violates admissibility completeness unless an additional gate excluding $W_{\text{obs|adm}} \setminus U$ is added to Γ .*

Corollary 6.8 (Conditional uniqueness of $W_{\text{obs|adm}}$). *The final admissible sector is unique conditional on the complete protocol but is not universal with respect to changes of protocol.*

Theorem 6.9 (Natural transport of fixed-protocol geometry). *Let $f : (\mathcal{E}, \Gamma) \rightarrow (\mathcal{E}', \Gamma')$ be an experiment isomorphism transporting the complete protocol. Then f induces an isomorphism of quotient domains, transports the Fisher tensor covariantly, conjugates the hard projector, and maps the maximal admissible sector onto the maximal admissible sector:*

$$f_* W_{\text{obs|adm}} = W'_{\text{obs|adm}}. \quad (18)$$

Proof. Quotient naturality follows from equality-of-law preservation. Metric covariance is part of the metric module. Spectral functional calculus is equivariant under conjugation. Since the complete gate family is transported, $f_* W_{\text{obs|adm}}$ is admissible in the target. For any target-admissible U' , its inverse image is source-admissible and hence lies in $W_{\text{obs|adm}}$; therefore $U' \subseteq f_* W_{\text{obs|adm}}$. Thus the image is target-maximal and equals the unique target sector. $\square$

Theorem 6.10 (Fixed-protocol observable-geometry representation). *Let $(\mathcal{E}, \Gamma)$ be a regular finite-dimensional classical declared admissible experiment. Assume:*

1. *observational extensionality and quotient universality;*
2. *the regular classical Fisher metric module with declared normalization;*
3. *a positive comparison metric G , a threshold τ separated from the generalized spectrum, and the hard spectral-resolution axioms;*
4. *a gate family admitting a greatest admissible resolved subobject;*
5. *admissibility soundness and completeness;*
6. *natural transport, identity, cocycle gluing, and independent-product composition.*

Then the observable geometry

$$\text{RG}(\mathcal{E}, \Gamma) = (Q_{\mathcal{E}}, g_F, P_{\tau}, W_{\text{obs|adm}}, \{T_{ij}\}, \Gamma) \quad (19)$$

is determined uniquely up to protocol-preserving natural isomorphism and the declared Fisher normalization.

Proof. The quotient representation theorem uniquely fixes $Q_{\mathcal{E}}$ up to unique compatible isomorphism. The imported metric characterization fixes g_F up to its declared normalization. The spectral projector theorem uniquely fixes P_{τ} . The maximal admissible-sector theorem uniquely fixes $W_{\text{obs|adm}}$. Naturality uniquely determines transported copies from these fixed local data, while identity and cocycle compatibility make direct and composite identifications agree on overlaps. Product composition excludes a distinct assignment on independent sums. Therefore two realizations of the same bundled input are connected componentwise by protocol-preserving isomorphisms. The quotient map and transported structural maps determine the compatible isomorphism; the componentwise naturality squares commute. Hence the realizations are naturally isomorphic. $\square$

Remark 6.11 (Exact force of established literature). *The quotient and metric layers inherit genuine established characterization theorems. The projector layer is a direct consequence of the spectral theorem and the fixed threshold mask. The admissible-sector layer is a greatest-subobject argument in a subspace lattice. Category theory then supplies the correct uniqueness language: universal or greatest constructions are unique only through structure-preserving maps. The full RG theorem is therefore modular; no cited classical theorem, by itself, states the assembled RG result.*

Corollary 6.12 (Fibrewise canonicity). *For each fixed admissibility protocol Γ , the RG assignment defines a canonical functor*

$$\text{RG}_\Gamma : \text{Exp}_\Gamma \longrightarrow \text{ObsGeom}_\Gamma \quad (20)$$

within the theorem's domain. There is no asserted canonical identification between fibres belonging to different protocols.

Corollary 6.13 (No canonical protocol-free section). *The representation theorem does not provide a map $\mathcal{E} \mapsto \Gamma$. Consequently, it does not define a canonical section of the projection from declared experiments to bare experiments.*

7 Entropy as a quotient corollary

Entropy is not an axiom of RG. It is a downstream construction once a probability measure is placed on a microstate space and pushed through an observational quotient.

Theorem 7.1 (Observable-Quotient Entropy Decomposition). *Let X be a discrete microstate variable and $Q = \pi(X)$ a deterministic observational quotient. Then*

$$H(X) = H(Q) + H(X \mid Q). \quad (21)$$

If $S_G = k_B H(X)$, then

$$\boxed{S_G = k_B H(Q) + k_B H(X \mid Q)}. \quad (22)$$

Proof. Since Q is a deterministic function of X , $H(X, Q) = H(X)$. Apply the Shannon chain rule. $\square$

Corollary 7.2 (Boltzmann entropy in quotient fibers). *If $X \mid Q = q$ is uniform on a fiber of multiplicity $\Omega(q)$, then*

$$H(X \mid Q = q) = \log \Omega(q) \quad (23)$$

and

$$S_G = k_B H(Q) + \mathcal{E}_Q[k_B \log \Omega(Q)]. \quad (24)$$

Corollary 7.3 (Equality boundary). *The quotient Shannon entropy equals total Gibbs entropy in units of k_B if and only if*

$$H(X \mid Q) = 0, \quad (25)$$

equivalently, π is injective on $\text{supp } p$.

Corollary 7.4 (Resolution redistribution). *If Q_f refines Q_c , then*

$$H(Q_f) \geq H(Q_c), \quad H(X \mid Q_f) \leq H(X \mid Q_c), \quad (26)$$

while $H(X)$ remains fixed. Increased resolution transfers entropy from unresolved fibers to observable classes; it does not create total entropy.

8 Finite fixed-protocol attack

8.1 Design

A fixed six-mode protocol was declared in a G -orthonormal information eigenbasis:

$$\lambda = (0.18, 0.42, 0.91, 1.37, 2.15, 3.80), \quad \tau = 1. \quad (27)$$

The nuisance, stability, conditioning, coarse-graining, and transport masks were frozen before enumeration. The canonical resolved and admissible masks were

$$P_\tau : (0, 0, 0, 1, 1, 1), \quad (28)$$

$$W_{\text{obs|adm}} : (0, 0, 0, 1, 0, 0). \quad (29)$$

Every binary hard spectral projector mask and every binary final-sector mask were paired, giving

$$2^6 \times 2^6 = 4096 \quad (30)$$

competitors.

8.2 Elimination ledger

Table 2: Survivors in the exhaustive fixed-protocol attack.

Constraint	Survivors
Initial hard spectral/sector pairs	4096
Fixed-threshold consistency	64
Gate soundness	2
Gate completeness/maximality	1
Transport naturality	1
Product composition	1

Exactly one pair survived, equal to the canonical construction. Soundness alone left two competitors: the canonical one-dimensional sector and its zero strict subspace. Completeness removed the zero sector. Thus the finite attack did not merely fail to find alternatives; it isolated an explicit independence witness for the new axiom.

A second attack generated 128 random oblique similarities of the canonical projection. None simultaneously satisfied idempotence, metric self-adjointness, and commutation with the information operator.

Proposition 8.1 (Finite battery result). *Within the enumerated six-mode hard-spectral class, the fixed protocol admits exactly one observable-sector assignment satisfying threshold consistency, gate soundness and completeness, natural transport, and product composition.*

Proof. Exhaustive enumeration checks all 4096 binary spectral/projected-sector pairs. The survivor ledger above is generated deterministically by the accompanying script. The unique surviving masks coincide with the analytic construction. $\square$

Remark 8.2. *The computation is not a proof of the representation theorem. The theorem follows from the modular arguments above. The battery instead checks implementation, exhibits the necessity of completeness in the finite model, and searches the full stated binary competitor class for overlooked alternatives.*

9 Axiom roles and independence

Proposition 9.1 (Soundness does not imply completeness). *There exist fixed protocols for which more than one sound sector exists.*

Proof. If the canonical maximal sector is nonzero, both it and the zero subspace satisfy containment in every gate. Only the maximal sector satisfies completeness. $\square$

Proposition 9.2 (Completeness does not select a protocol). *Admissibility completeness is conditional on Γ and does not derive G , τ , N , S , C , or T from $\mathcal{E}$.*

Proof. Different protocols define different admissible families $\text{Adm}_\Gamma(R_\tau)$ and therefore may have different greatest elements. Completeness selects the greatest object within a declared family; it does not select the family. $\square$

Proposition 9.3 (Threshold consistency is independent of hard projection). *Idempotence, metric self-adjointness, and commutation with L admit every orthogonal spectral subprojection. Threshold consistency is required to select the specified upper spectral set.*

Proof. In a spectral basis, any binary choice on eigenspaces defines a commuting orthogonal projector. Only the threshold mask fixes which eigenspaces are retained. $\square$

10 Tautological closure and exact locus of choice

Theorem 10.1 (Fibrewise closure theorem). *Within the regular finite classical domain and for a fixed bundled input $(\mathcal{E}, \Gamma)$ satisfying the stated existence hypotheses, every component of $\text{RG}(\mathcal{E}, \Gamma)$ is either (i) forced by an established universal, characterization, spectral, greatest-subobject, or gluing property, or (ii) explicitly present in Γ . No third, undeclared choice remains.*

Proof. The quotient is forced by equality-of-law extensionality and its universal property. The metric is forced, up to declared normalization, by the imported classical metric characterization. The hard projector is forced by spectral functional calculus and threshold consistency. The admissible sector is forced by soundness plus completeness as the greatest common admissible subobject. Transported copies and overlap identifications are forced relative to the declared morphisms by naturality and cocycle closure. All remaining choices—comparison scale, threshold, nuisance convention, gate class, and transition data—are components of Γ and are therefore visible assumptions rather than hidden degrees of freedom. $\square$

Corollary 10.2 (Exact tautology statement). *RG is mathematically tautological only in the disciplined sense that its output is the uniquely structured consequence of its declared input and axioms. It is not vacuous: changing or violating an input axiom changes, destroys, or renders non-unique the resulting object.*

Non-claim 10.3 (No derivation of the contract). *The closure theorem does not derive the observation system or protocol from nothing. It proves absence of undeclared freedom after they are specified.*

11 Interpretation

The representation theorem identifies RG with neither a bare quotient nor a universal final sector. The quotient is universal at the domain layer. The final sector is maximal only inside a fixed protocol fibre. The complete logical order is

$$(\mathcal{E}, \Gamma) \longrightarrow Q_{\mathcal{E}} \longrightarrow g_F \longrightarrow P_\tau \longrightarrow W_{\text{obs|adm}} \longrightarrow \{T_{ij}\}. \quad (31)$$

The word “canonical” must therefore be read fibrewise. A protocol is to RG what a metric is to Riemannian geometry: declared structure whose consequences are characterized. This is weaker than protocol-free uniqueness and stronger than presenting an arbitrary construction. It says that no additional undeclared choice remains after the complete admissibility contract and the RG axioms are fixed.

The theorem schema is modular. A future quantum version could replace the classical Fisher module with a characterized quantum distinguishability module while retaining quotient extensionality, hard or graded spectral resolution, admissibility maximality, and natural transport, provided the appropriate categories and subobject meets exist. No such extension is proved here.

12 Non-claims and falsification criteria

Non-claim 12.1 (No protocol-free uniqueness). *The paper does not assert that $\mathcal{E}$ determines Γ , or that all observation geometries are RG .*

Non-claim 12.2 (No universality of $W_{\text{obs}|\text{adm}}$). *The universal object is the equality-of-law quotient for experiment-internal maps. The final admissible sector is a maximal subobject relative to a fixed protocol, not an initial or terminal object across all protocols.*

Non-claim 12.3 (No unrestricted Fisher theorem). *Fisher–Rao is imported only in the regular finite classical domain under the stated invariance and normalization assumptions. Singular, infinite-dimensional, non-dominated, and quantum experiments require separate metric modules.*

Non-claim 12.4 (No automatic global uniqueness). *Atlas uniqueness is relative to declared transition data satisfying identity and cocycle laws. Different undeclared connections or holonomy classes are different protocols, not rival realizations in one fibre.*

Non-claim 12.5 (No physical ontology). *The quotient expresses operational indistinguishability. It neither denies latent representatives nor identifies the observable geometry with spacetime, quantum state reality, or any physical ontology.*

Non-claim 12.6 (No infinite-dimensional maximality without hypotheses). *In general categories or topological vector spaces, arbitrary gate intersections may fail to belong to the intended regularity class or may require closure. The finite linear greatest-subobject proof does not automatically extend.*

Falsification and revision conditions. The stated characterization fails or requires revision if:

1. an experiment-internal quantity varies inside an equality-of-law class;
2. a metric satisfying the stated classical module is not proportional to Fisher–Rao;
3. two distinct hard projectors satisfy all fixed (F, G, τ) selector axioms off the spectrum;
4. two distinct greatest admissible subobjects exist in the same fixed gate family;
5. covariant transport changes the characterized object beyond natural isomorphism;
6. direct and composite overlap transports disagree;
7. a claimed stable sector has no positive margin from the declared discriminant.

13 Open problems

1. Prove existence of greatest admissible subobjects for nonlinear, stratified, and sheaf-valued protocol categories.
2. Determine the correct closure operation for infinite-dimensional gate intersections and establish perturbation-stable maximality.
3. Extend the representation theorem to singular and rank-changing experiments.
4. Formulate a quantum experiment category and replace the classical metric module by an explicitly characterized monotone quantum module.
5. Separate protocol-preserving natural equivalence from equivalence under protocol refinement or Blackwell comparison.
6. Determine when decision, resource, or physical axioms select a subfamily of protocols; this is external to the present fibrewise theorem.

14 Conclusion

The correct classical identification is now precise. Resolution Geometry is the canonical observable geometry of a declared admissible experiment, not the unique protocol-free geometry of observation. Quotient universality fixes the observable domain. The classical metric characterization fixes Fisher–Rao distinguishability in its domain. Spectral theory fixes the hard resolved projector. The newly explicit completeness axiom fixes the final sector as the greatest resolved subobject satisfying every declared gate. Naturality and cocycle laws fix transport and gluing relative to the contract.

Thus the defensible uniqueness statement is fibrewise:

$$(\mathcal{E}, \Gamma) \implies \text{RG}(\mathcal{E}, \Gamma) \quad \text{up to protocol-preserving natural isomorphism.} \quad (32)$$

The finite battery supports this statement, exposes completeness as necessary, and finds no competing hard-spectral realization in the exhaustive class. Protocol selection remains outside the theorem by design.

Reproducibility

The deterministic supporting artifacts are:

- `rg_fixed_protocol_uniqueness_attack.py`;
- `rg_fixed_protocol_uniqueness_attack.json`.

The seed is 20260805. The exhaustive binary enumeration itself is seed-independent; the seed controls the supplementary oblique-projector attack.

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A Battery pseudocode

```

fix information spectrum, threshold, and all protocol gates
canonical_P = spectral_mask(information > threshold)
canonical_W = canonical_P AND all_gate_masks

for P in all binary spectral projectors:
  for W in all binary sector masks:
    require P == canonical_P           # threshold consistency
    require W subset_of (P AND all_gates) # soundness
    require W == (P AND all_gates)      # completeness
    require covariance under transported protocol
    require direct-sum composition
    retain(P,W)

assert exactly one retained pair
assert retained pair == (canonical_P, canonical_W)

```

B Machine-readable result summary

all competitors:	4096
survive threshold consistency:	64
survive gate soundness:	2
survive gate completeness:	1
survive natural transport:	1
survive product composition:	1
random oblique projectors tested:	128
oblique survivors of all projector rules:	0